

JESS Thermodynamic Database v8.9

Technetium

24-Jan-23

Reaction No. 42685							
$\text{TcO}+2\text{_{OH-1}} + \text{NTA-3} = \text{TcO}+2\text{_{OH-1_NTA-3}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	NaClO ₄	lgK:13.8 (0.2SD)	2	MIX	4350
2	25	0	Inf. Dilution	lgK:14.0 (3SF)	1	ESC	

Reaction No. 42686							
$\text{TcO}+2\text{_{OH-1}} + 2\text{<NTA-3>} = \text{TcO}+2\text{_{OH-1_NTA-3(2)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	NaClO ₄	lgK:25.7 (0.2SD)	0	MIX	4350
2	25	0	Inf. Dilution	lgK:30.0 (3SF)	1	ESC	

Reaction No. 42687							
$\text{TcO}+2\text{_{OH-1}} + \text{CDTA-4} = \text{TcO}+2\text{_{OH-1_CDTA-4}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.035	Unknown	lgK:20.4 (0.2SD)	5	MTP	4350
2	23	0.1	NaClO ₄	lgK:20.7 (0.2SD)	5	MIX	4350
3	25	0	Inf. Dilution	lgK:20.7 (3SF)	1	ESC	

Reaction No. 42688							
$\text{TcO}+2\text{_{OH-1}} + \text{EDTA-4} = \text{TcO}+2\text{_{OH-1_EDTA-4}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.1	NaClO ₄	lgK:19.1 (0.2SD)	3	MTP	4350
2	25	0	Inf. Dilution	lgK:19.4 (3SF)	1	ESC	
3	25	0.1	Unknown	lgK:19.1 (3SF)	2	EOD	29557

Reaction No. 54416							
$\text{Tc}+4\text{_{Br-1(5)}} + \text{Br-1} = \text{Tc}+4\text{_{Br-1(6)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	15	3	HClO ₄	lgK:3.58 (3SF)	4	MAG	931

Reaction No. 55878							
$\text{TcO}_4\text{-1} + \text{e-1} = \text{TcO}_4\text{-2}$							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.80(4SF)	4	CRV	32222
2	25	0	CHEMVAL1	lgK:-10.32(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-10.32(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-10.32(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-10.30(0.01SD)	3	ENS	5156

Reaction No. 55879							
TcO4-1 + 2<e-1> = TcO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:-20.29(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL2	lgK:-20.29(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL3	lgK:-20.29(0.01SD)	4	ENS	5156
4	25	0	CHEMVAL4	lgK:-20.30(0.01SD)	4	ENS	5156

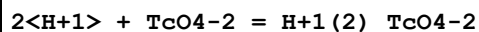
Reaction No. 55880							
TcO4-1 + 6<H+1> + 3<e-1> = TcO+2 + 3<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:33.45(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL2	lgK:33.45(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL3	lgK:33.45(0.01SD)	4	ENS	5156
4	25	0	CHEMVAL4	lgK:33.40(0.01SD)	4	ENS	5156

Reaction No. 55881							
TcO4-1 + 8<H+1> + 4<e-1> = Tc+3 + 4<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:38.83(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL2	lgK:38.83(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL3	lgK:38.83(0.01SD)	4	ENS	5156
4	25	0	CHEMVAL4	lgK:38.80(0.01SD)	4	ENS	5156

Reaction No. 55882							
H+1 + TcO4-2 = H+1_TcO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:-1.61(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL2	lgK:-1.61(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL3	lgK:-1.61(0.01SD)	4	ENS	5156

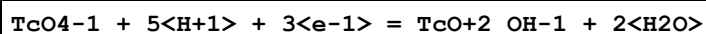
4	25	0	CHEMVAL4	lgK:-1.61(0.01SD)	4	ENS	5156
---	----	---	----------	-------------------	---	-----	------

Reaction No. 55883



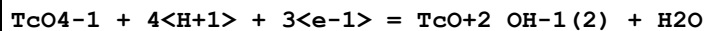
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:-1.31(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL2	lgK:-1.31(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL3	lgK:-1.31(0.01SD)	4	ENS	5156
4	25	0	CHEMVAL4	lgK:-1.31(0.01SD)	4	ENS	5156

Reaction No. 55884



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:34.6(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:32.32(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:32.32(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:32.32(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:32.30(0.01SD)	0	ENS	5156

Reaction No. 55885



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:30.14(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL2	lgK:30.14(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL3	lgK:30.14(0.01SD)	4	ENS	5156
4	25	0	CHEMVAL4	lgK:29.90(0.01SD)	4	ENS	5156

Reaction No. 56128



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-40.6(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-41.30(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-41.30(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-41.30(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-41.30(0.01SD)	0	ENS	5156
6	25	0	Inf. Dilution	E:-0.782(3SF)	0	CRV	6330
7	25	0	Inf. Dilution	E:-0.782(3SF)	0	CRV	22519

Reaction No. 56129							
$\text{TcO}_2 \cdot 2\text{H}_2\text{O}(\text{s}) = \text{TcO}_4^- + 4\text{H}^+ + 3\text{e}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	$\lg K: -37.70 (0.01\text{SD})$	0	ENS	5156
2	25	0	CHEMVAL2	$\lg K: -37.70 (0.01\text{SD})$	0	ENS	5156
3	25	0	CHEMVAL3	$\lg K: -37.70 (0.01\text{SD})$	0	ENS	5156
4	25	0	CHEMVAL4	$\lg K: -37.70 (0.01\text{SD})$	4	ENS	5156

Reaction No. 56130							
$\text{Tc} + 3\text{OH}^- + \text{H}_2\text{O} = \text{TcO}_4^- + 5\text{H}^+ + 4\text{e}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	$\lg K: -48.10 (0.01\text{SD})$	4	ENS	5156
2	25	0	CHEMVAL2	$\lg K: -48.10 (0.01\text{SD})$	4	ENS	5156
3	25	0	CHEMVAL3	$\lg K: -48.10 (0.01\text{SD})$	4	ENS	5156
4	25	0	CHEMVAL4	$\lg K: -48.10 (0.01\text{SD})$	4	ENS	5156

Reaction No. 56131							
$\text{TcO}_3(\text{s}) + \text{H}_2\text{O} = \text{TcO}_4^- + 2\text{H}^+ + \text{e}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	$\lg K: -13.50 (0.01\text{SD})$	4	ENS	5156
2	25	0	CHEMVAL2	$\lg K: -13.50 (0.01\text{SD})$	4	ENS	5156
3	25	0	CHEMVAL3	$\lg K: -13.50 (0.01\text{SD})$	4	ENS	5156
4	25	0	CHEMVAL4	$\lg K: -13.50 (0.01\text{SD})$	4	ENS	5156

Reaction No. 56132							
$\text{Tc}_2\text{O}_7(\text{s}) + 8\text{H}_2\text{O} = 3\text{TcO}_4^- + 16\text{H}^+ + 13\text{e}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	$\lg K: -157.30 (0.01\text{SD})$	4	ENS	5156
2	25	0	CHEMVAL2	$\lg K: -157.30 (0.01\text{SD})$	4	ENS	5156
3	25	0	CHEMVAL3	$\lg K: -157.30 (0.01\text{SD})$	4	ENS	5156
4	25	0	CHEMVAL4	$\lg K: -157.00 (0.01\text{SD})$	4	ENS	5156

Reaction No. 56133							
$\text{Tc}_2\text{O}_7(\text{s}) + 9\text{H}_2\text{O} = 4\text{TcO}_4^- + 18\text{H}^+ + 14\text{e}^-$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	$\lg K: -170.60 (0.01\text{SD})$	4	ENS	5156

2	25	0	CHEMVAL2	lgK:-170.60(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL3	lgK:-170.60(0.01SD)	4	ENS	5156
4	25	0	CHEMVAL4	lgK:-171.00(0.01SD)	4	ENS	5156

Reaction No. 56134							
K+1_TcO4-1_(s) = K+1 + TcO4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.28(1.28SD)	4	CRV	32222
2	25	0	CHEMVAL1	lgK:-2.73(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-2.73(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-2.73(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-2.60(0.01SD)	3	ENS	5156
6	25	0	Inf. Dilution	dH:53.560(10.748SD)kJ	4	CRV	32222

Reaction No. 56135							
Na+1_TcO4-1_(s) = Na+1 + TcO4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:1.05(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL2	lgK:1.05(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL3	lgK:1.05(0.01SD)	4	ENS	5156
4	25	0	CHEMVAL4	lgK:1.05(0.01SD)	4	ENS	5156

Reaction No. 56136							
H+1_TcO4-1_(s) = H+1 + TcO4-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:5.50(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL2	lgK:5.50(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL3	lgK:5.50(0.01SD)	4	ENS	5156
4	25	0	CHEMVAL4	lgK:5.50(0.01SD)	4	ENS	5156

Reaction No. 56137							
Tc+4_S-2(2)_(s) + 4<H2O> = TcO4-1 + 2<H+1_S-2> + 6<H+1> + 3<e-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:-99.45(0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:-99.45(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-99.45(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-99.40(0.01SD)	2	ENS	5156

Reaction No. 56138							
$\text{Tc}+6_S-2(3)_ (s) + 4\text{<H}_2\text{O>} = \text{TcO}_4-1 + 3\text{<H}+1_S-2> + 5\text{<H}+1> + e-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:-109.80(0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:-109.80(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-109.80(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-110.00(0.01SD)	3	ENS	5156

Reaction No. 56139							
$\text{Tc}_2\text{S}_7(s) + 8\text{<H}_2\text{O>} = 2\text{<TcO}_4-1> + 7\text{<H}+1_S-2> + 9\text{<H}+1>$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:-231.40(0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:-231.40(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:-231.40(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:-231.00(0.01SD)	2	ENS	5156

Reaction No. 56140							
$\text{Tc}(s) + 4\text{<H}_2\text{O>} = \text{TcO}_4-1 + 8\text{<H}+1> + 7\text{<e}^{-1}>$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-57.4(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-57.37(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-57.37(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-57.37(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-57.40(0.01SD)	0	ENS	5156
6	25	0	Inf. Dilution	E:-0.472(3SF)	0	CRV	6330
7	25	0	Inf. Dilution	E:-0.472(3SF)	0	CRV	22519

Reaction No. 56294							
$\text{Tc}_2\text{O}_7(s) + \text{H}_2\text{O} = 2\text{<TcO}_4-1> + 2\text{<H}+1>$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.8(3SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:15.31(3.82SD)	0	CRV	32222
3	25	0	CHEMVAL1	lgK:12.17(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:12.17(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:12.17(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:12.20(0.01SD)	0	ENS	5156

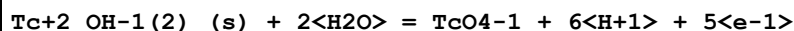
7	25	0	Inf. Dilution	dH:-46.470 (21.285SD)kJ	4	CRV	32222
---	----	---	---------------	-------------------------	---	-----	-------

Reaction No. 56301



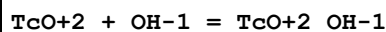
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL4	lgK:-56.90 (0.01SD)	4	ENS	5156

Reaction No. 56327



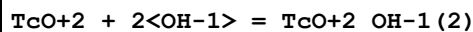
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:-55.08 (0.01SD)	4	ENS	5156
2	25	0	CHEMVAL2	lgK:-55.08 (0.01SD)	4	ENS	5156
3	25	0	CHEMVAL3	lgK:-55.08 (0.01SD)	4	ENS	5156
4	25	0	CHEMVAL4	lgK:-55.10 (0.01SD)	4	ENS	5156

Reaction No. 60508



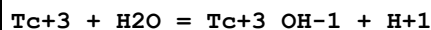
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:12.6 (3SF)	5	CRV	2556

Reaction No. 60509



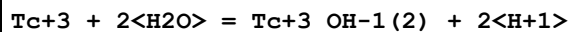
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:24.2 (3SF)	5	CRV	2556

Reaction No. 64968



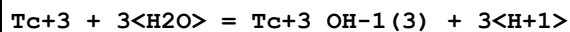
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.23 (3SF)	1	EBU	6372

Reaction No. 64969



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.61 (3SF)	1	EBU	6372

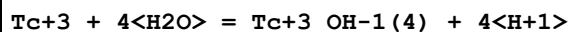
Reaction No. 64970



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
----	---	-------	--------	-------------	---	-------	--------

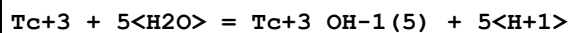
1	25	0	Inf. Dilution	lgK:-16.02(4SF)	1	EBU	6372
---	----	---	---------------	-----------------	---	-----	------

Reaction No. 64971



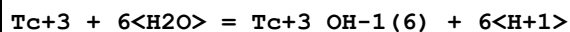
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-23.40(4SF)	1	EBU	6372

Reaction No. 64972



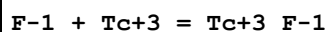
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-31.73(4SF)	1	EBU	6372

Reaction No. 64973



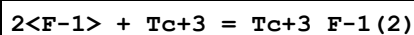
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-41.00(4SF)	1	EBU	6372

Reaction No. 64974



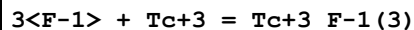
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.11(3SF)	1	EBU	6372

Reaction No. 64975



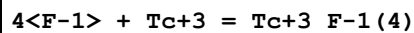
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.55(4SF)	1	EBU	6372

Reaction No. 64976



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.44(4SF)	1	EBU	6372

Reaction No. 64977



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.85(4SF)	1	EBU	6372

Reaction No. 64978

5<F-1> + Tc+3 = Tc+3_F-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.78(4SF)	1	EBU	6372

Reaction No. 64979							
6<F-1> + Tc+3 = Tc+3_F-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.28(4SF)	1	EBU	6372

Reaction No. 64980							
Cl-1 + Tc+3 = Tc+3_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.80(3SF)	1	EBU	6372

Reaction No. 64981							
2<Cl-1> + Tc+3 = Tc+3_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.14(3SF)	1	EBU	6372

Reaction No. 64982							
3<Cl-1> + Tc+3 = Tc+3_Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.14(3SF)	1	EBU	6372

Reaction No. 64983							
4<Cl-1> + Tc+3 = Tc+3_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.84(3SF)	1	EBU	6372

Reaction No. 64984							
5<Cl-1> + Tc+3 = Tc+3_Cl-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.29(3SF)	1	EBU	6372

Reaction No. 64985							
6<Cl-1> + Tc+3 = Tc+3_Cl-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.49(3SF)	1	EBU	6372

Reaction No. 64986							
Br-1 + Tc+3 = Tc+3_Br-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.53(3SF)	1	EBU	6372

Reaction No. 64987							
2<Br-1> + Tc+3 = Tc+3_Br-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.71(3SF)	1	EBU	6372

Reaction No. 64988							
3<Br-1> + Tc+3 = Tc+3_Br-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.67(3SF)	1	EBU	6372

Reaction No. 64989							
4<Br-1> + Tc+3 = Tc+3_Br-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.47(3SF)	1	EBU	6372

Reaction No. 64990							
5<Br-1> + Tc+3 = Tc+3_Br-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.12(3SF)	1	EBU	6372

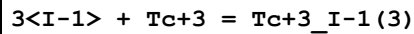
Reaction No. 64991							
6<Br-1> + Tc+3 = Tc+3_Br-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.65(3SF)	1	EBU	6372

Reaction No. 64992							
I-1 + Tc+3 = Tc+3_I-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.48(3SF)	1	EBU	6372

Reaction No. 64993							
2<I-1> + Tc+3 = Tc+3_I-1(2)							

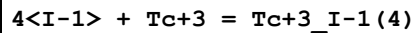
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.65 (3SF)	1	EBU	6372

Reaction No. 64994



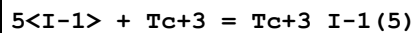
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.65 (3SF)	1	EBU	6372

Reaction No. 64995



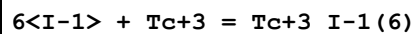
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.51 (3SF)	1	EBU	6372

Reaction No. 64996



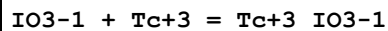
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.27 (3SF)	1	EBU	6372

Reaction No. 64997



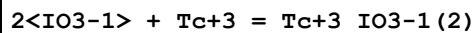
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.96 (3SF)	1	EBU	6372

Reaction No. 64998



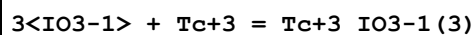
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.38 (3SF)	1	EBU	6372

Reaction No. 64999



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.72 (3SF)	1	EBU	6372

Reaction No. 65000



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.30 (3SF)	1	EBU	6372

Reaction No. 65001							
4<IO3-1> + Tc+3 = Tc+3_IO3-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.20 (3SF)	1	EBU	6372

Reaction No. 65002							
5<IO3-1> + Tc+3 = Tc+3_IO3-1 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.48 (3SF)	1	EBU	6372

Reaction No. 65003							
6<IO3-1> + Tc+3 = Tc+3_IO3-1 (6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.18 (3SF)	1	EBU	6372

Reaction No. 65004							
SO4-2 + Tc+3 = Tc+3_SO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.91 (3SF)	1	EBU	6372

Reaction No. 65005							
2<SO4-2> + Tc+3 = Tc+3_SO4-2 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.31 (3SF)	1	EBU	6372

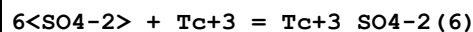
Reaction No. 65006							
3<SO4-2> + Tc+3 = Tc+3_SO4-2 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.44 (3SF)	1	EBU	6372

Reaction No. 65007							
4<SO4-2> + Tc+3 = Tc+3_SO4-2 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.40 (3SF)	1	EBU	6372

Reaction No. 65008							
5<SO4-2> + Tc+3 = Tc+3_SO4-2 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

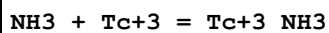
1	25	0	Inf. Dilution	lgK:6.26(3SF)	1	EBU	6372
---	----	---	---------------	---------------	---	-----	------

Reaction No. 65009



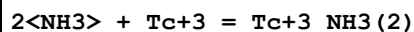
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.05(3SF)	1	EBU	6372

Reaction No. 65010



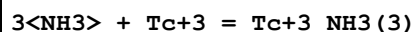
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.39(3SF)	1	EBU	6372

Reaction No. 65011



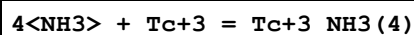
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.76(4SF)	1	EBU	6372

Reaction No. 65012



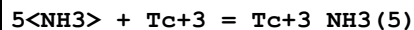
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.22(4SF)	1	EBU	6372

Reaction No. 65013



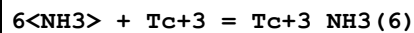
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.84(4SF)	1	EBU	6372

Reaction No. 65014



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.64(4SF)	1	EBU	6372

Reaction No. 65015



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.64(4SF)	1	EBU	6372

Reaction No. 65016

NO3-1 + Tc+3 = Tc+3_NO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.37(3SF)	1	EBU	6372

Reaction No. 65017							
2<NO3-1> + Tc+3 = Tc+3_NO3-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.47(3SF)	1	EBU	6372

Reaction No. 65018							
3<NO3-1> + Tc+3 = Tc+3_NO3-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.55(2SF)	1	EBU	6372

Reaction No. 65019							
4<NO3-1> + Tc+3 = Tc+3_NO3-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.28(3SF)	1	EBU	6372

Reaction No. 65020							
5<NO3-1> + Tc+3 = Tc+3_NO3-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.97(3SF)	1	EBU	6372

Reaction No. 65021							
Tc+3 + H+1_PO4-3 = Tc+3_H+1_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.81(3SF)	1	EBU	6372

Reaction No. 65022							
Tc+3 + 2<H+1_PO4-3> = Tc+3_H+1(2)_PO4-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.50(4SF)	1	EBU	6372

Reaction No. 65023							
Tc+3 + 3<H+1_PO4-3> = Tc+3_H+1(3)_PO4-3(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.33(4SF)	1	EBU	6372

Reaction No. 65024							
$\text{Tc}+3 + 4\text{<H+1_PO4-3>} = \text{Tc}+3\text{_H+1(4)_PO4-3(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.39(4SF)	1	EBU	6372

Reaction No. 65025							
$\text{Tc}+3 + 5\text{<H+1_PO4-3>} = \text{Tc}+3\text{_H+1(5)_PO4-3(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:39.74(4SF)	1	EBU	6372

Reaction No. 65026							
$\text{Tc}+3 + 6\text{<H+1_PO4-3>} = \text{Tc}+3\text{_H+1(6)_PO4-3(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:45.42(4SF)	1	EBU	6372

Reaction No. 65027							
$\text{Tc}+3 + \text{H+1(2)_PO4-3} = \text{Tc}+3\text{_H+1(2)_PO4-3}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.52(3SF)	1	EBU	6372

Reaction No. 65028							
$\text{Tc}+3 + 2\text{<H+1(2)_PO4-3>} = \text{Tc}+3\text{_H+1(4)_PO4-3(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.85(3SF)	1	EBU	6372

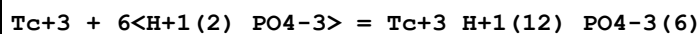
Reaction No. 65029							
$\text{Tc}+3 + 3\text{<H+1(2)_PO4-3>} = \text{Tc}+3\text{_H+1(6)_PO4-3(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.22(3SF)	1	EBU	6372

Reaction No. 65030							
$\text{Tc}+3 + 4\text{<H+1(2)_PO4-3>} = \text{Tc}+3\text{_H+1(8)_PO4-3(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.75(3SF)	1	EBU	6372

Reaction No. 65031							
$\text{Tc}+3 + 5\text{<H+1(2)_PO4-3>} = \text{Tc}+3\text{_H+1(10)_PO4-3(5)}$							

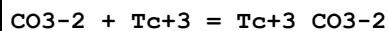
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.48 (3SF)	1	EBU	6372

Reaction No. 65032



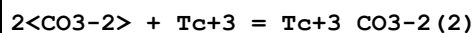
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.46 (3SF)	1	EBU	6372

Reaction No. 65033



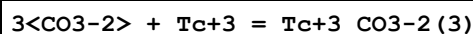
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.23 (3SF)	1	EBU	6372

Reaction No. 65034



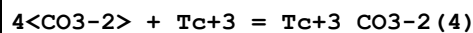
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.92 (4SF)	1	EBU	6372

Reaction No. 65035



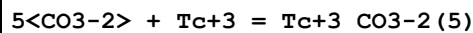
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.32 (4SF)	1	EBU	6372

Reaction No. 65036



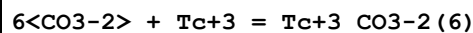
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.54 (4SF)	1	EBU	6372

Reaction No. 65037



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.63 (4SF)	1	EBU	6372

Reaction No. 65038



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:35.63 (4SF)	1	EBU	6372

Reaction No. 65039							
$\text{Tc}+3 + \text{H}+1_ \text{CO}3-2 = \text{Tc}+3_ \text{H}+1_ \text{CO}3-2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.05 (3SF)	1	EBU	6372

Reaction No. 65040							
$\text{Tc}+3 + 2<\text{H}+1_ \text{CO}3-2> = \text{Tc}+3_ \text{H}+1(2)_ \text{CO}3-2(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.50 (3SF)	1	EBU	6372

Reaction No. 65041							
$\text{Tc}+3 + 3<\text{H}+1_ \text{CO}3-2> = \text{Tc}+3_ \text{H}+1(3)_ \text{CO}3-2(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.59 (4SF)	1	EBU	6372

Reaction No. 65042							
$\text{Tc}+3 + 4<\text{H}+1_ \text{CO}3-2> = \text{Tc}+3_ \text{H}+1(4)_ \text{CO}3-2(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.44 (4SF)	1	EBU	6372

Reaction No. 65043							
$\text{Tc}+3 + 5<\text{H}+1_ \text{CO}3-2> = \text{Tc}+3_ \text{H}+1(5)_ \text{CO}3-2(5)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.09 (4SF)	1	EBU	6372

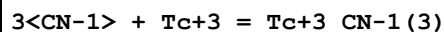
Reaction No. 65044							
$\text{Tc}+3 + 6<\text{H}+1_ \text{CO}3-2> = \text{Tc}+3_ \text{H}+1(6)_ \text{CO}3-2(6)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.59 (3SF)	1	EBU	6372

Reaction No. 65045							
$\text{CN}-1 + \text{Tc}+3 = \text{Tc}+3_ \text{CN}-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.45 (3SF)	1	EBU	6372

Reaction No. 65046							
$2<\text{CN}-1> + \text{Tc}+3 = \text{Tc}+3_ \text{CN}-1(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

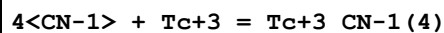
1	25	0	Inf. Dilution	lgK:13.81 (4SF)	1	EBU	6372
---	----	---	---------------	-----------------	---	-----	------

Reaction No. 65047



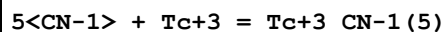
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.22 (4SF)	1	EBU	6372

Reaction No. 65048



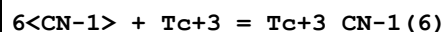
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.73 (4SF)	1	EBU	6372

Reaction No. 65049



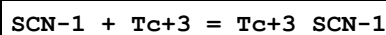
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.36 (4SF)	1	EBU	6372

Reaction No. 65050



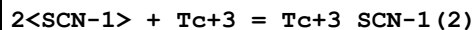
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:30.14 (4SF)	1	EBU	6372

Reaction No. 65051



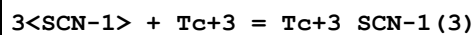
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.19 (3SF)	1	EBU	6372

Reaction No. 65052



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.45 (3SF)	1	EBU	6372

Reaction No. 65053



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.90 (3SF)	1	EBU	6372

Reaction No. 65054

4<SCN-1> + Tc+3 = Tc+3_SCN-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.59(3SF)	1	EBU	6372

Reaction No. 65055							
5<SCN-1> + Tc+3 = Tc+3_SCN-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.54(3SF)	1	EBU	6372

Reaction No. 65056							
6<SCN-1> + Tc+3 = Tc+3_SCN-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.79(2SF)	1	EBU	6372

Reaction No. 65057							
Acetic-1 + Tc+3 = Tc+3_Acetic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.28(3SF)	1	EBU	6372

Reaction No. 65058							
2<Acetic-1> + Tc+3 = Tc+3_Acetic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.97(3SF)	1	EBU	6372

Reaction No. 65059							
3<Acetic-1> + Tc+3 = Tc+3_Acetic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.33(3SF)	1	EBU	6372

Reaction No. 65060							
4<Acetic-1> + Tc+3 = Tc+3_Acetic-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.45(3SF)	1	EBU	6372

Reaction No. 65061							
5<Acetic-1> + Tc+3 = Tc+3_Acetic-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.39(3SF)	1	EBU	6372

Reaction No. 65062							
6<Acetic-1> + Tc+3 = Tc+3_Acetic-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.19(3SF)	1	EBU	6372

Reaction No. 65063							
Benzoic-1 + Tc+3 = Tc+3_Benzoic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.77(3SF)	1	EBU	6372

Reaction No. 65064							
2<Benzoic-1> + Tc+3 = Tc+3_Benzoic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.04(3SF)	1	EBU	6372

Reaction No. 65065							
3<Benzoic-1> + Tc+3 = Tc+3_Benzoic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.05(4SF)	1	EBU	6372

Reaction No. 65066							
4<Benzoic-1> + Tc+3 = Tc+3_Benzoic-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.91(4SF)	1	EBU	6372

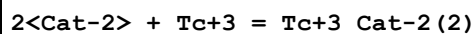
Reaction No. 65067							
5<Benzoic-1> + Tc+3 = Tc+3_Benzoic-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.67(4SF)	1	EBU	6372

Reaction No. 65068							
6<Benzoic-1> + Tc+3 = Tc+3_Benzoic-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.38(3SF)	1	EBU	6372

Reaction No. 65069							
Cat-2 + Tc+3 = Tc+3_Cat-2							

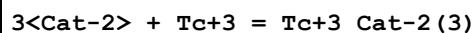
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.33 (4SF)	1	EBU	6372

Reaction No. 65070



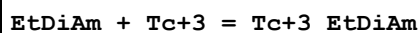
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.64 (4SF)	1	EBU	6372

Reaction No. 65071



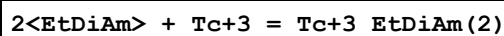
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:55.60 (4SF)	1	EBU	6372

Reaction No. 65072



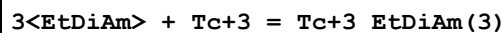
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.23 (4SF)	1	EBU	6372

Reaction No. 65073



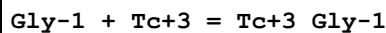
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.67 (4SF)	1	EBU	6372

Reaction No. 65074



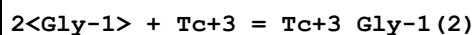
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.00 (4SF)	1	EBU	6372

Reaction No. 65075



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.78 (3SF)	1	EBU	6372

Reaction No. 65076



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.77 (4SF)	1	EBU	6372

Reaction No. 65077							
3<Gly-1> + Tc+3 = Tc+3_Gly-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.00(4SF)	1	EBU	6372

Reaction No. 65078							
Glycolic-1 + Tc+3 = Tc+3_Glycolic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.84(3SF)	1	EBU	6372

Reaction No. 65079							
2<Glycolic-1> + Tc+3 = Tc+3_Glycolic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.07(3SF)	1	EBU	6372

Reaction No. 65080							
3<Glycolic-1> + Tc+3 = Tc+3_Glycolic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.96(3SF)	1	EBU	6372

Reaction No. 65081							
4<Glycolic-1> + Tc+3 = Tc+3_Glycolic-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.59(3SF)	1	EBU	6372

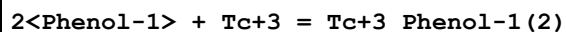
Reaction No. 65082							
5<Glycolic-1> + Tc+3 = Tc+3_Glycolic-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.04(3SF)	1	EBU	6372

Reaction No. 65083							
6<Glycolic-1> + Tc+3 = Tc+3_Glycolic-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.33(3SF)	1	EBU	6372

Reaction No. 65084							
Phenol-1 + Tc+3 = Tc+3_Phenol-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

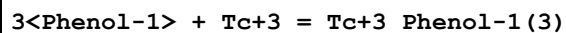
1	25	0	Inf. Dilution	lgK:7.22 (3SF)	1	EBU	6372
---	----	---	---------------	----------------	---	-----	------

Reaction No. 65085



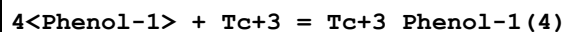
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.19 (4SF)	1	EBU	6372

Reaction No. 65086



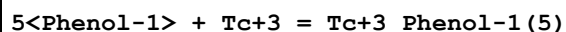
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.01 (4SF)	1	EBU	6372

Reaction No. 65087



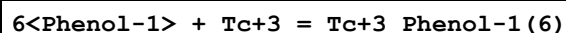
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.75 (4SF)	1	EBU	6372

Reaction No. 65088



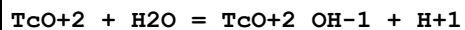
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.43 (4SF)	1	EBU	6372

Reaction No. 65089



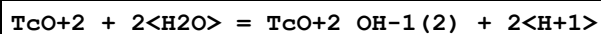
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.07 (4SF)	1	EBU	6372

Reaction No. 65090



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Inf. Dilution	lgK:1.39 (3SF)	4	CRV	32224
2	25	0	Inf. Dilution	lgK:1.2 (2SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:-1.67 (3SF)	0	EBU	6372
4	25	0	Inf. Dilution	lgK:0.0072 (2SF)	0	EOD	29882

Reaction No. 65091



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
----	---	-------	--------	-------------	---	-------	--------

1	20	0.1	Inf. Dilution	lgK:-3.78 (3SF)	4	CRV	32224
2	25	0	Inf. Dilution	lgK:-3.4 (2SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:-4.08 (3SF)	0	EBU	6372
4	25	0	Inf. Dilution	lgK:-4. (1SF)	0	EOD	29882

Reaction No. 65092							
$\text{TcO}+2 + 3\langle\text{H}_2\text{O}\rangle = \text{TcO}+2_{\text{OH}-1(3)} + 3\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.12 (3SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:-14.9 (3SF)	3	EOD	29882

Reaction No. 65093							
$\text{TcO}+2 + 4\langle\text{H}_2\text{O}\rangle = \text{TcO}+2_{\text{OH}-1(4)} + 4\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.72 (4SF)	1	EBU	6372

Reaction No. 65094							
$\text{TcO}+2 + 5\langle\text{H}_2\text{O}\rangle = \text{TcO}+2_{\text{OH}-1(5)} + 5\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-14.87 (4SF)	1	EBU	6372

Reaction No. 65095							
$\text{F}-1 + \text{TcO}+2 = \text{TcO}+2_{\text{F}-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.70 (3SF)	1	EBU	6372

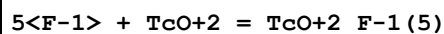
Reaction No. 65096							
$2\langle\text{F}-1\rangle + \text{TcO}+2 = \text{TcO}+2_{\text{F}-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.33 (4SF)	1	EBU	6372

Reaction No. 65097							
$3\langle\text{F}-1\rangle + \text{TcO}+2 = \text{TcO}+2_{\text{F}-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.00 (4SF)	1	EBU	6372

Reaction No. 65098							
$4\langle\text{F}-1\rangle + \text{TcO}+2 = \text{TcO}+2_{\text{F}-1(4)}$							

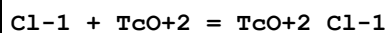
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.76(4SF)	1	EBU	6372

Reaction No. 65099



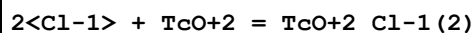
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.65(4SF)	1	EBU	6372

Reaction No. 65100



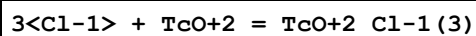
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.55(3SF)	1	EBU	6372

Reaction No. 65101



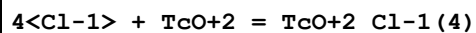
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.76(3SF)	1	EBU	6372

Reaction No. 65102



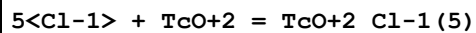
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.77(3SF)	1	EBU	6372

Reaction No. 65103



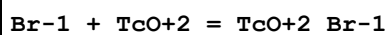
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.62(3SF)	1	EBU	6372

Reaction No. 65104



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.33(3SF)	1	EBU	6372

Reaction No. 65105



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.37(3SF)	1	EBU	6372

Reaction No. 65106							
$2\langle\text{Br-1}\rangle + \text{TcO}+2 = \text{TcO}+2_{\text{Br-1}}(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.43(3SF)	1	EBU	6372

Reaction No. 65107							
$3\langle\text{Br-1}\rangle + \text{TcO}+2 = \text{TcO}+2_{\text{Br-1}}(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.31(3SF)	1	EBU	6372

Reaction No. 65108							
$4\langle\text{Br-1}\rangle + \text{TcO}+2 = \text{TcO}+2_{\text{Br-1}}(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.06(3SF)	1	EBU	6372

Reaction No. 65109							
$5\langle\text{Br-1}\rangle + \text{TcO}+2 = \text{TcO}+2_{\text{Br-1}}(5)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.71(3SF)	1	EBU	6372

Reaction No. 65110							
$\text{I-1} + \text{TcO}+2 = \text{TcO}+2_{\text{I-1}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.39(3SF)	1	EBU	6372

Reaction No. 65111							
$2\langle\text{I-1}\rangle + \text{TcO}+2 = \text{TcO}+2_{\text{I-1}}(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.47(3SF)	1	EBU	6372

Reaction No. 65112							
$3\langle\text{I-1}\rangle + \text{TcO}+2 = \text{TcO}+2_{\text{I-1}}(3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.38(3SF)	1	EBU	6372

Reaction No. 65113							
$4\langle\text{I-1}\rangle + \text{TcO}+2 = \text{TcO}+2_{\text{I-1}}(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:4.17 (3SF)	1	EBU	6372
---	----	---	---------------	----------------	---	-----	------

Reaction No. 65114							
5<I-1> + TcO+2 = TcO+2_I-1 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.86 (3SF)	1	EBU	6372

Reaction No. 65115							
IO3-1 + TcO+2 = TcO+2_IO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.02 (3SF)	1	EBU	6372

Reaction No. 65116							
2<IO3-1> + TcO+2 = TcO+2_IO3-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.31 (3SF)	1	EBU	6372

Reaction No. 65117							
3<IO3-1> + TcO+2 = TcO+2_IO3-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.12 (3SF)	1	EBU	6372

Reaction No. 65118							
4<IO3-1> + TcO+2 = TcO+2_IO3-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.55 (3SF)	1	EBU	6372

Reaction No. 65119							
5<IO3-1> + TcO+2 = TcO+2_IO3-1 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.65 (3SF)	1	EBU	6372

Reaction No. 65120							
SO4-2 + TcO+2 = TcO+2_SO4-2							
Note: See analysis and commentary in Parker et al. [18POD_31363]							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.60 (3SF)	1	EBU	6372

Reaction No. 65121							
$2\text{<SO}_4\text{-2>} + \text{TcO}_2 = \text{TcO}_2\text{-SO}_4\text{-2 (2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.63(3SF)	1	EBU	6372

Reaction No. 65122							
$3\text{<SO}_4\text{-2>} + \text{TcO}_2 = \text{TcO}_2\text{-SO}_4\text{-2 (3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.35(3SF)	1	EBU	6372

Reaction No. 65123							
$4\text{<SO}_4\text{-2>} + \text{TcO}_2 = \text{TcO}_2\text{-SO}_4\text{-2 (4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.86(3SF)	1	EBU	6372

Reaction No. 65124							
$5\text{<SO}_4\text{-2>} + \text{TcO}_2 = \text{TcO}_2\text{-SO}_4\text{-2 (5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.21(3SF)	1	EBU	6372

Reaction No. 65125							
$\text{NH}_3 + \text{TcO}_2 = \text{TcO}_2\text{-NH}_3$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.16(3SF)	1	EBU	6372

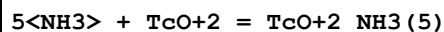
Reaction No. 65126							
$2\text{<NH}_3> + \text{TcO}_2 = \text{TcO}_2\text{-NH}_3\text{ (2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.71(4SF)	1	EBU	6372

Reaction No. 65127							
$3\text{<NH}_3> + \text{TcO}_2 = \text{TcO}_2\text{-NH}_3\text{ (3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.75(4SF)	1	EBU	6372

Reaction No. 65128							
$4\text{<NH}_3> + \text{TcO}_2 = \text{TcO}_2\text{-NH}_3\text{ (4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

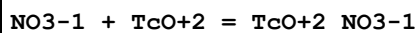
1	25	0	Inf. Dilution	lgK:25.36 (4SF)	1	EBU	6372
---	----	---	---------------	-----------------	---	-----	------

Reaction No. 65129



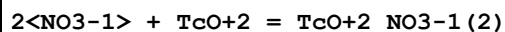
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:30.54 (4SF)	1	EBU	6372

Reaction No. 65130



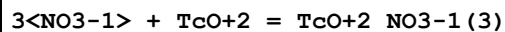
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.07 (3SF)	1	EBU	6372

Reaction No. 65131



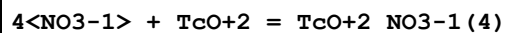
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.27 (3SF)	1	EBU	6372

Reaction No. 65132



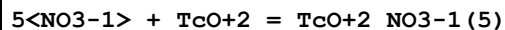
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.85 (2SF)	1	EBU	6372

Reaction No. 65133



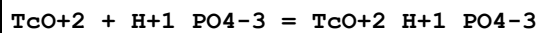
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.10 (2SF)	1	EBU	6372

Reaction No. 65134



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.52 (3SF)	1	EBU	6372

Reaction No. 65135



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.48 (4SF)	1	EBU	6372

Reaction No. 65136

TcO+2 + 2<H+1_PO4-3> = TcO+2_H+1(2)_PO4-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.46(4SF)	1	EBU	6372

Reaction No. 65137							
TcO+2 + 3<H+1_PO4-3> = TcO+2_H+1(3)_PO4-3(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.18(4SF)	1	EBU	6372

Reaction No. 65138							
TcO+2 + 4<H+1_PO4-3> = TcO+2_H+1(4)_PO4-3(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.75(4SF)	1	EBU	6372

Reaction No. 65139							
TcO+2 + 5<H+1_PO4-3> = TcO+2_H+1(5)_PO4-3(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:39.21(4SF)	1	EBU	6372

Reaction No. 65140							
TcO+2 + H+1(2)_PO4-3 = TcO+2_H+1(2)_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.17(3SF)	1	EBU	6372

Reaction No. 65141							
TcO+2 + 2<H+1(2)_PO4-3> = TcO+2_H+1(4)_PO4-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.79(3SF)	1	EBU	6372

Reaction No. 65142							
TcO+2 + 3<H+1(2)_PO4-3> = TcO+2_H+1(6)_PO4-3(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.10(3SF)	1	EBU	6372

Reaction No. 65143							
TcO+2 + 4<H+1(2)_PO4-3> = TcO+2_H+1(8)_PO4-3(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.21(3SF)	1	EBU	6372

Reaction No. 65144							
$\text{TcO}_2 + 5\text{H}^+ (2)_{\text{PO4-3}} = \text{TcO}_2\text{H}^+ (10)_{\text{PO4-3}} (5)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.17 (3SF)	1	EBU	6372

Reaction No. 65145							
$\text{CO}_3^{2-} + \text{TcO}_2 = \text{TcO}_2\text{CO}_3^{2-}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.57 (4SF)	1	EBU	6372

Reaction No. 65146							
$2\text{CO}_3^{2-} + \text{TcO}_2 = \text{TcO}_2\text{CO}_3^{2-} (2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.97 (4SF)	1	EBU	6372

Reaction No. 65147							
$3\text{CO}_3^{2-} + \text{TcO}_2 = \text{TcO}_2\text{CO}_3^{2-} (3)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.46 (4SF)	1	EBU	6372

Reaction No. 65148							
$4\text{CO}_3^{2-} + \text{TcO}_2 = \text{TcO}_2\text{CO}_3^{2-} (4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:36.13 (4SF)	1	EBU	6372

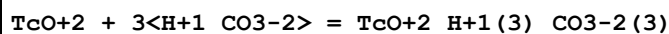
Reaction No. 65149							
$5\text{CO}_3^{2-} + \text{TcO}_2 = \text{TcO}_2\text{CO}_3^{2-} (5)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:43.05 (4SF)	1	EBU	6372

Reaction No. 65150							
$\text{TcO}_2 + \text{H}^+_{\text{CO}_3^{2-}} = \text{TcO}_2\text{H}^+_{\text{CO}_3^{2-}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.30 (3SF)	1	EBU	6372

Reaction No. 65151							
$\text{TcO}_2 + 2\text{H}^+_{\text{CO}_3^{2-}} = \text{TcO}_2\text{H}^+ (2)_{\text{CO}_3^{2-}} (2)$							

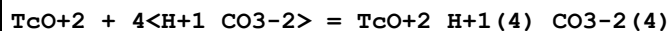
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.24(3SF)	1	EBU	6372

Reaction No. 65152



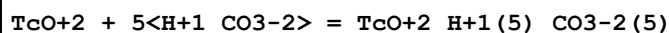
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.06(4SF)	1	EBU	6372

Reaction No. 65153



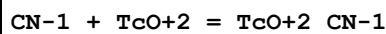
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.89(4SF)	1	EBU	6372

Reaction No. 65154



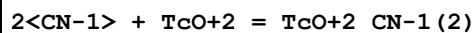
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.76(4SF)	1	EBU	6372

Reaction No. 65155



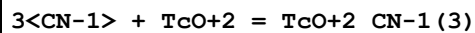
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.21(3SF)	1	EBU	6372

Reaction No. 65156



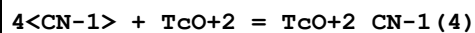
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.76(4SF)	1	EBU	6372

Reaction No. 65157



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.76(4SF)	1	EBU	6372

Reaction No. 65158



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.27(4SF)	1	EBU	6372

Reaction No. 65159							
5<CN-1> + TcO+2 = TcO+2_CN-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:35.32(4SF)	1	EBU	6372

Reaction No. 65160							
SCN-1 + TcO+2 = TcO+2_SCN-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.80(3SF)	1	EBU	6372

Reaction No. 65161							
2<SCN-1> + TcO+2 = TcO+2_SCN-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.06(3SF)	1	EBU	6372

Reaction No. 65162							
3<SCN-1> + TcO+2 = TcO+2_SCN-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.88(3SF)	1	EBU	6372

Reaction No. 65163							
4<SCN-1> + TcO+2 = TcO+2_SCN-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.33(3SF)	1	EBU	6372

Reaction No. 65164							
5<SCN-1> + TcO+2 = TcO+2_SCN-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.43(3SF)	1	EBU	6372

Reaction No. 65165							
Acetic-1 + TcO+2 = TcO+2_Acetic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.24(3SF)	1	EBU	6372

Reaction No. 65166							
2<Acetic-1> + TcO+2 = TcO+2_Acetic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

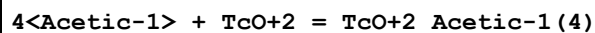
1	25	0	Inf. Dilution	lgK:7.21(3SF)	1	EBU	6372
---	----	---	---------------	---------------	---	-----	------

Reaction No. 65167



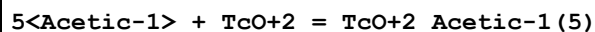
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.14(3SF)	1	EBU	6372

Reaction No. 65168



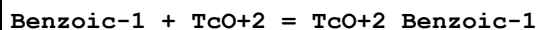
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.14(4SF)	1	EBU	6372

Reaction No. 65169



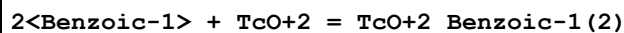
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.28(4SF)	1	EBU	6372

Reaction No. 65170



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.91(3SF)	1	EBU	6372

Reaction No. 65171



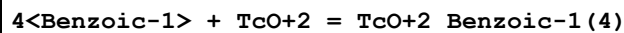
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.70(3SF)	1	EBU	6372

Reaction No. 65172



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.63(4SF)	1	EBU	6372

Reaction No. 65173



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.81(4SF)	1	EBU	6372

Reaction No. 65174

5<Benzoic-1> + TcO+2 = TcO+2_Benzoic-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.27(4SF)	1	EBU	6372

Reaction No. 65175							
Cat-2 + TcO+2 = TcO+2_Cat-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.40(4SF)	1	EBU	6372

Reaction No. 65176							
2<Cat-2> + TcO+2 = TcO+2_Cat-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:47.31(4SF)	1	EBU	6372

Reaction No. 65177							
EtDiAm + TcO+2 = TcO+2_EtDiAm							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.22(4SF)	1	EBU	6372

Reaction No. 65178							
2<EtDiAm> + TcO+2 = TcO+2_EtDiAm(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:31.17(4SF)	1	EBU	6372

Reaction No. 65179							
Gly-1 + TcO+2 = TcO+2_Gly-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.42(3SF)	1	EBU	6372

Reaction No. 65180							
2<Gly-1> + TcO+2 = TcO+2_Gly-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.24(4SF)	1	EBU	6372

Reaction No. 65181							
Glycolic-1 + TcO+2 = TcO+2_Glycolic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.68(3SF)	1	EBU	6372

Reaction No. 65182							
2<Glycolic-1> + TcO+2 = TcO+2_Glycolic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.00(3SF)	1	EBU	6372

Reaction No. 65183							
3<Glycolic-1> + TcO+2 = TcO+2_Glycolic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.23(3SF)	1	EBU	6372

Reaction No. 65184							
4<Glycolic-1> + TcO+2 = TcO+2_Glycolic-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.45(3SF)	1	EBU	6372

Reaction No. 65185							
5<Glycolic-1> + TcO+2 = TcO+2_Glycolic-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.74(3SF)	1	EBU	6372

Reaction No. 65186							
Phenol-1 + TcO+2 = TcO+2_Phenol-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.23(3SF)	1	EBU	6372

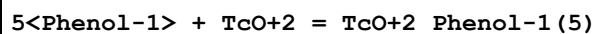
Reaction No. 65187							
2<Phenol-1> + TcO+2 = TcO+2_Phenol-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.54(4SF)	1	EBU	6372

Reaction No. 65188							
3<Phenol-1> + TcO+2 = TcO+2_Phenol-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.06(4SF)	1	EBU	6372

Reaction No. 65189							
4<Phenol-1> + TcO+2 = TcO+2_Phenol-1(4)							

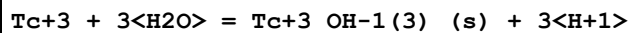
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.85 (4SF)	1	EBU	6372

Reaction No. 65190



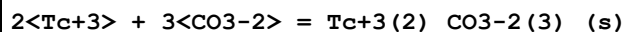
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.92 (4SF)	1	EBU	6372

Reaction No. 65510



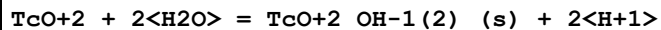
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.3 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-5.90 (3SF)	0	EBU	6372

Reaction No. 65511



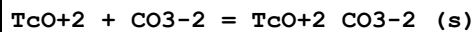
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.93 (4SF)	1	EBU	6372

Reaction No. 65512



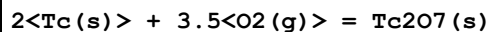
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.78 (3SF)	1	EBU	6372

Reaction No. 65513



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.20 (4SF)	1	EBU	6372

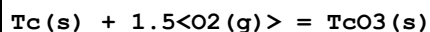
Reaction No. 66371



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:164.3 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:163.1 (4SF)	4	ENS	6422
3	120	0	Inf. Dilution	lgK:117.3 (4SF)	4	ENS	6422
4	25	0	Inf. Dilution	dG:-224.1 (4SF)	4	EFE	6422
5	25	0	Inf. Dilution	dH:-266.4 (4SF)	4	MTD	6422

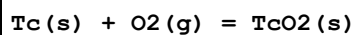
6	120	0	Inf. Dilution	dH:-264.2 (4SF)	4	MTD	6422
---	-----	---	---------------	-----------------	---	-----	------

Reaction No. 66372



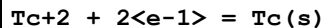
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:80.78 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:80.20 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:56.79 (4SF)	4	ENS	6422
4	25	0	Inf. Dilution	dG:-110.2 (4SF)	4	EFE	6422
5	25	0	Inf. Dilution	dH:-129.0 (4SF)	4	MTD	6422
6	127	0	Inf. Dilution	dH:-127.9 (4SF)	4	MTD	6422

Reaction No. 66373



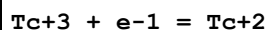
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:66.24 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:65.78 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:46.94 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:35.66 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:28.15 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-90.37 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-103.5 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-103.4 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-103.1 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-102.8 (4SF)	4	MTD	6422

Reaction No. 70986



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.400 (3SF)	5	CRV	6330
2	25	0	Inf. Dilution	E:0.400 (3SF)	4	CRV	22519

Reaction No. 70987



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.3 (1SF)	5	CRV	6330
2	25	0	Inf. Dilution	E:0.3 (1SF)	4	CRV	22519

Reaction No. 72377							
$2\text{<O2(g)>} + \text{Tc(s)} + \text{e-1} = \text{TcO4-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:108.8(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-36.(2SF)kJ	0	CRV	7421
3	25	0	Inf. Dilution	dG:-151.100(6SF)	0	CRV	9029
4	25	0	Inf. Dilution	dH:-173.200(6SF)	2	CRV	9029

Reaction No. 79671							
$2\text{<H+1_TcO4-1_(s)>} = \text{Tc2O7(s)} + \text{H2O(g)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:3.686(0.018SD)	5	ETR	29294
2	25	0	Inf. Dilution	dH:10.950(0.050SD)	5	ETR	29294

Reaction No. 79854							
$\text{TcO2.2H2O(s)} = \text{TcO+2} + 2\text{<OH-1>} + \text{H2O}$							
Note: Probably refers to $\text{TcO2.2H2O(am.,s)} = \text{TcO+2} + 2\text{<OH-1>} + \text{H2O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-32.3(3SF)	1	ELC	

Reaction No. 79855							
$\text{TcO2.2H2O(am.,s)} = \text{TcO+2} + 2\text{<OH-1>} + \text{H2O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-33.6(0.32SD)	3	EOD	29557

Reaction No. 79856							
$\text{TcO+2} + \text{EDTA-4} = \text{TcO+2_EDTA-4}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.2(0.6SD)	2	MSL	29557
Note (1): Ambiguity over NDP and ionic strength!							

Reaction No. 79857							
$\text{TcO+2} + \text{Picolinic-1} = \text{TcO+2_Picolinic-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.9(0.1SD)	2	MSL	29557
Note (1): Ambiguity over corresponding EDTA NDP and ionic strength!							

Reaction No. 79879							
$2\text{<S(s)>} + \text{Tc(s)} = \text{Tc+4_S-2(2)_(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-215.9(0.042SD)kJ	3	EOD	29614
Note (1): p.27: Est x McDonald & Cobble, 1962							
2	25	0	Inf. Dilution	dH:-223.8(0.041SD)kJ	3	EOD	29614

Reaction No. 79880							
$7\text{<S(s)>} + 2\text{<Tc(s)>} = \text{Tc2S7(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-580.5(0.060SD)kJ	3	EOD	29614
2	25	0	Inf. Dilution	dH:-615.0(0.057SD)kJ	3	EOD	29614

Reaction No. 80023							
$\text{DTPA-5} + \text{TcO+2} = \text{TcO+2_DTPA-5}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaCl	lgK:20.1(0.1MD)	1	MDS	29882
Note (1): Analogous VO+2 complex suspect							
2	25	0.5	NaNO3	lgK:22.0(0.06MD)	0	EOD	29882
Note (2): Footnote claims this is for B111							

Reaction No. 80024							
$\text{HOEDTA-3} + \text{TcO+2} = \text{TcO+2_HOEDTA-3}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaCl	lgK:14.9(0.1MD)	2	MDS	29882
2	25	0.5	NaNO3	lgK:15.3(0.3MD)	2	EOD	29882

Reaction No. 80025							
$\text{NTA-3} + \text{TcO+2} = \text{TcO+2_NTA-3}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaCl	lgK:11.4(0.1MD)	1	MDS	29882
2	25	0.5	NaNO3	lgK:10.3(0.3MD)	1	EOD	29882

Reaction No. 80026							
$\text{IDA-2} + \text{TcO+2} = \text{TcO+2_IDA-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaCl	lgK:10.9(0.1MD)	0	MDS	29882

Note (1): Less accord with VO+2 spc							
2	25	0.5	NaNO3	lgK:9.2(0.3MD)	2	EOD	29882

Reaction No. 80617							
TcO+2_OH-1(2) + H+1 = TcO+2_OH-1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.563(0.11SD)	5	CRV	31301
2	25	0.8	NaCl	lgK:4.551(0.15SD)	5	CRV	31301
3	25	2.5	NaCl	lgK:4.552(0.29SD)	5	CRV	31301

Reaction No. 80618							
TcO2,1.6H2O(s) = TcO+2_OH-1(2) + 0.6<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.415(0.09SD)	5	CRV	31301
2	25	0.8	NaCl	lgK:-8.620(0.14SD)	5	CRV	31301
3	25	2.5	NaCl	lgK:-8.210(0.17SD)	5	CRV	31301

Reaction No. 80626							
Oxalic-2 + TcO+2 = TcO+2_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.50(3SF)	4	CRV	31301

Reaction No. 80627							
2<Oxalic-2> + TcO+2 = TcO+2_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.21(4SF)	4	CRV	31301

Reaction No. 80630							
Citric-3 + TcO+2 = TcO+2_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.99(4SF)	4	CRV	31301

Reaction No. 80631							
TcO+2 + H+1_Citric-3 = TcO+2_H+1_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.75(3SF)	4	CRV	31301

Reaction No. 80689							
--------------------	--	--	--	--	--	--	--

TcO+2_OH-1 + SO4-2 = TcO+2_OH-1_SO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	NaCl	lgK:1.13(0.04SD)	4	MSE	31363
Note (1): unspecified temp							
2	25	0	Inf. Dilution	lgK:1.5(2SF)	1	ESC	31363

Reaction No. 80727							
0.5<O2(g)> + Tc(s) - 2<e-1> = TcO+2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-100.60(5SF)kJ	3	EOD	31585

Reaction No. 80728							
2<O2(g)> + Tc(s) + 2<e-1> = TcO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-564.90(5SF)kJ	3	EOD	31585

Reaction No. 80729							
2<O2(g)> + Tc(s) + 3<e-1> = TcO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-508.00(5SF)kJ	3	EOD	31585

Reaction No. 80731							
Tc(s) - 3<e-1> = Tc+3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:105.80(5SF)kJ	3	EOD	31585

Reaction No. 81035							
Tc2O7.H2O(s) = 2<TcO4-1> + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.11(3.81SD)	4	CRV	32222
2	25	0	Inf. Dilution	dH:-44.654(21.288SD)kJ	4	CRV	32222

Reaction No. 81036							
TcO+2_OH-1(2) = TcO2(s) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.5(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:12.39(2.58SD)	0	CRV	32222

Reaction No. 81037							
$\text{TcO}_4\text{-1} + \text{Na+1} + 4\text{<H}_2\text{O>} = \text{NaTcO}_4.4\text{H}_2\text{O(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -0.79(0.04\text{SD})$	4	CRV	32222

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
EBU	Brown's Unified Theory
EFE	Free Energy relations
ELC	Linear Combination of Reactions
ENS	Not specified
EOD	Calculated from data of others
ESC	Standard conditions anchor
ETR	Thermodynamic relationships
MAG	Silver electrode e.m.f.
MDS	Distribution between two phases
MIX	Ion exchange

MSE	Solvent Extraction
MSL	Solubility
MTD	Temperature difference
MTP	Migration or transference number

References

Reference No.: 931(71SiM)

Stability Constants of Metal-ion Complexes SP25, 1971 Sillen LG; Martell AE; Supplement No 1 to Special Publication No 25; Chemical Society, London

Reference No.: 2556(93MSM)

NIST Critical Stability Constants of Metal Complexes Database, 1993 Martell AE; Smith RM; Motekaitis RJ; Version 1.0; National Institute of Standards and Technology, Gaithersburg, USA

Reference No.: 4350(70GoK)

J. Inorg. Nucl. Chem., 1970, 32, 3831 - 3836; Gorski B; Koch H; Uber Die Komplexbildung Von Technetium Mit Chelatbildenden Liganden II

Reference No.: 5156(91ReB)

Radiochim. Acta, 1991, 52/53, 453 - 456; Read D; Broyd TW; Recent Progress in Testing Chemical Equilibrium Models: The CHEMVAL Project

Reference No.: 6330(95LiF)

CRC Handbook of Chemistry and Physics, 76th Edn., 1995 Lide DR; Frederikse HPR; A Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio, U.S.A.

Reference No.: 6372(87BrW)

OECD NEA Report, 1987 Brown PL; Wanner H; Predicted Formation Constants Using the Unified Theory of Metal Ion Complexation; Nuclear Energy Authority, Paris

Reference No.: 6422(95BaP)

Thermochemical Data of Pure Substances, 3rd Edn., 1995 Barin I; Platzki G; Weinheim, Germany

Reference No.: 7421(97Mar)

Ion Properties, 1997 Marcus Y; Marcel Dekker Inc., New York, U.S.A.

Reference No.: 9029(97SSW)

Geochim. Cosmochim. Acta, 1997, 61, 907 - 950; Shock EL; Sassani DC; Willis M; Sverjensky DA; Inorganic Species in Geologic Fluids: Correlations among Standard Molal Thermodynamic Properties of Aqueous Ions and Hydroxide Complexes

Reference No.: 22519(05Lid)

CRC Handbook of Chemistry and Physics, 86th Edn., 2005 Lide DR; A Ready-Reference Book of Chemical and Physical Data; Taylor & Francis, Boca Raton, USA

Reference No.: 29294(53SCB)

J. Am. Chem. Soc., 1953, 75, 5773 - 5776; Smith WT Jr; Cobble JW; Boyd GE;

Thermodynamic Properties of Technetium and Rhenium Compounds. I. Vapor Pressures of Technetium Heptoxide, Pertechnic Acid and Aqueous Solutions of Pertechnic Acid

Reference No.: 29557(12EvH)

Miner. Mag., 2012, 76, 3435 - 3438; Evans NDM; Hallam R; The Complexation of Tc(IV) with EDTA and Picolinic Acid at High pH

Reference No.: 29614(89Rar)

Report UCRL-100554, 1989 Rard JA; Chemical Thermodynamics of Technetium; Lawrence Livermore Natl. Lab., Livermore, CA, USA

Reference No.: 29882(17OmW)

J. Solution Chem., 2017, 46, 1981 - 1994; Omoto T; Wall NA; Evaluation of Vanadium(IV) as a Non-radioactive Surrogate for Technetium(IV) by Comparison of Stability Constants for Polyamino Polycarboxylate Ligand Complexation

Reference No.: 31301(10KFD)

Japan Atomic Energy Agency Data/Code 2009-024, 2010 Kitamura A; Fujiwara K; Doi R; Yoshida Y; Mihara M; Terashima M; Yui M; JAEA Thermodynamic Database for Performance Assessment of Geological Disposal of High-level Radioactive and TRU Wastes; Geological Isolation Research Unit, JAEA, Japan

Reference No.: 31363(18POD)

J. Solution Chem., 2018, 47, 1192 - 1201; Parker TG; Omoto T; Dickens SM; Wall DE; Wall NA; Complexation of Tc(IV) with SO₄²⁻ in NaCl Medium

Reference No.: 31585(19RVE)

J. Hazard. Mater., 2019, 380, 120725-1 - 120725-17; Ram R; Vaughan J; Etschmann B; Brugger J; The Aqueous Chemistry of Polonium (Po) in Environmental and Anthropogenic Processes

Reference No.: 32222(06DGC)

Technical Report TR-O6-17, 2006 Duro L; Grive M; Cera E; Domenech C; Bruno J; Update of a Thermodynamic Database for Radionuclides to Assist Solubility Limits Calculation for Performance Assessment; Swedish Nucl. Fuel & Waste Management Co, Stockholm, Sweden

Reference No.: 32224(80BeT)

Lawrence Berkeley Laboratory Report LBL-11448, 1980 Benson LV; Teague LS; A Tabulation of Thermodynamic Data for Chemical Reactions Involving 8 Elements Common to Radioactive Waste Package Systems